

MUKESH RAM B.

Python & AI Full-Stack Developer | Machine Learning & Backend Engineer

+91-8072543404 • mukeshramb1702@gmail.com • linkedin.com/in/mukeshramb1702 • Chennai, India

PROFESSIONAL SUMMARY

Associate Software Engineer with a degree in **Artificial Intelligence & Data Science (B.Tech)** and enterprise experience building data reconciliation modules, automated API verification suites, and high-performance deep learning pipelines. Expert in **Python, PyTorch, CUDA GPU Acceleration, FastAPI, OpenCV, and AWS**. Proficient in **AI-Assisted Software Development & AI Pair Programming** workflows (Copilot, Antigravity, Cursor) to accelerate prototyping, code optimization, and architectural debugging.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript

AI & Computer Vision: PyTorch, Real-ESRGAN (RRDBNet), OpenCV, CUDA FP16 Acceleration, NumPy

AI-Assisted Engineering: AI Pair Programming (Copilot, Antigravity, Cursor), Generative AI Tooling, LLM Prompting & API Integration (Gemini, OpenAI)

Backend & Web: FastAPI, Flask, REST APIs, WebSockets, Uvicorn, Asyncio, Node.js

Databases & Cloud: MySQL, PostgreSQL, MongoDB, Firebase, AWS (EC2, S3, IAM), Cloudflare Tunnels, Git, Linux

Testing & Automation: PyTest, API Verification, SwaggerHub, Bruno, Postman, Custom Python Automation

WORK EXPERIENCE

Apprentice Software Engineer — *Fidelity Investments*

Sept 2025 – Present

InTACT — Enterprise Data Reconciliation Platform

- Co-developed data reconciliation workflows integrating a high-throughput **Python comparison engine** with backend microservices across Oracle, Snowflake, CSV, XLSX, and JSON data sources.
- Built **Test Parameterization** with dynamic SQL query placeholder parsing, reducing redundant query executions and cutting dataset compare time by **35%** for large enterprise databases.
- Worked across UI, REST API layers, and Python compare execution engines to debug integration edge cases in Agile sprints; utilized **Git** and **Jira** for version control and sprint planning.

API Validation & Backend Verification Rotation

- Authored custom **Python verification scripts** and complex MySQL query pipelines to compare legacy vs. modernized system outputs, automating backend validation prior to production sign-off.
- Verified REST API contract compliance, status code handling, and payload integrity using **SwaggerHub** and **Bruno**.

FEATURED AI & SOFTWARE PROJECTS

Mukeus Video Enhancer — Local AI Video Processing Platform

PyTorch · CUDA FP16 · FastAPI · OpenCV · FFmpeg

- Designed and built a creator-focused AI video enhancement engine that upscales low-bitrate gaming footage and YouTube Shorts (16:9 & 9:16 vertical) to crisp 1080p/4K resolution using Real-ESRGAN (RRDBNet).
- Engineered an optimized **PyTorch inference pipeline** using **CUDA FP16 mixed precision** and **native GPU tensor interpolation**, achieving a **50x speedup** (0.03s per frame).
- Built an asynchronous **FastAPI backend** handling frame extraction, single-pass GPU execution, vector image filtering, and original audio remuxing via **FFmpeg**.
- Created a modern dark-themed web UI (HTML5, Vanilla CSS3, JS) featuring real-time frame progress tracking and side-by-side HTML5 video player comparison.

AstroGPT — AI-Powered Interpretation Engine

Python · Google Gemini AI · REST APIs · SSE · CI/CD

- Developed an AI analytics platform connecting custom calculation engines with **Google Gemini AI** via Server-Sent Events (SSE) for real-time generative analysis.
- Built automated test suites and CI/CD pipelines via GitHub Actions for continuous cloud deployment.

Sollidam — Anonymous Peer Listening Platform

Angular · Firebase Firestore · Jitsi Meet · Vercel

- Built a responsive full-stack booking web app with real-time slot management backed by Firebase Firestore.
- Handled Vercel serverless deployment with automated notifications, drawing **500+ impressions** on launch day.

EDUCATION & CERTIFICATIONS

B.Tech in Artificial Intelligence & Data Science (2020 – 2024)

Vel Tech Multi Tech Engineering College

AWS Certified Cloud Practitioner (CLF-C02)

Amazon Web Services